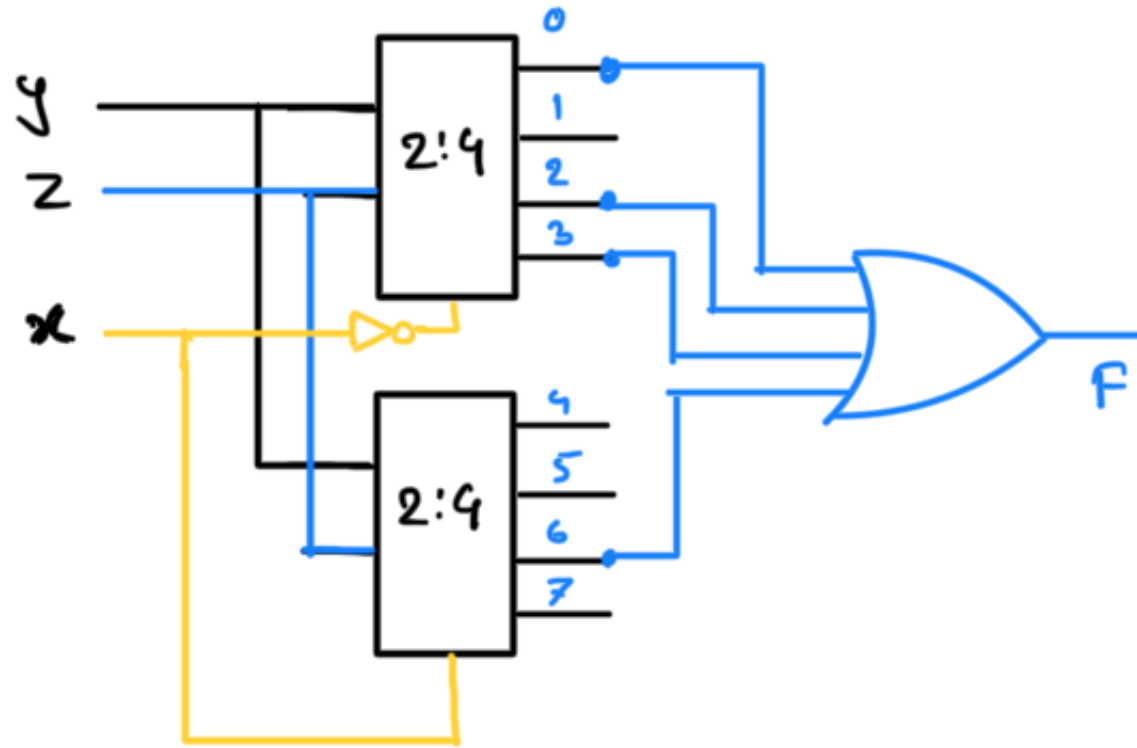


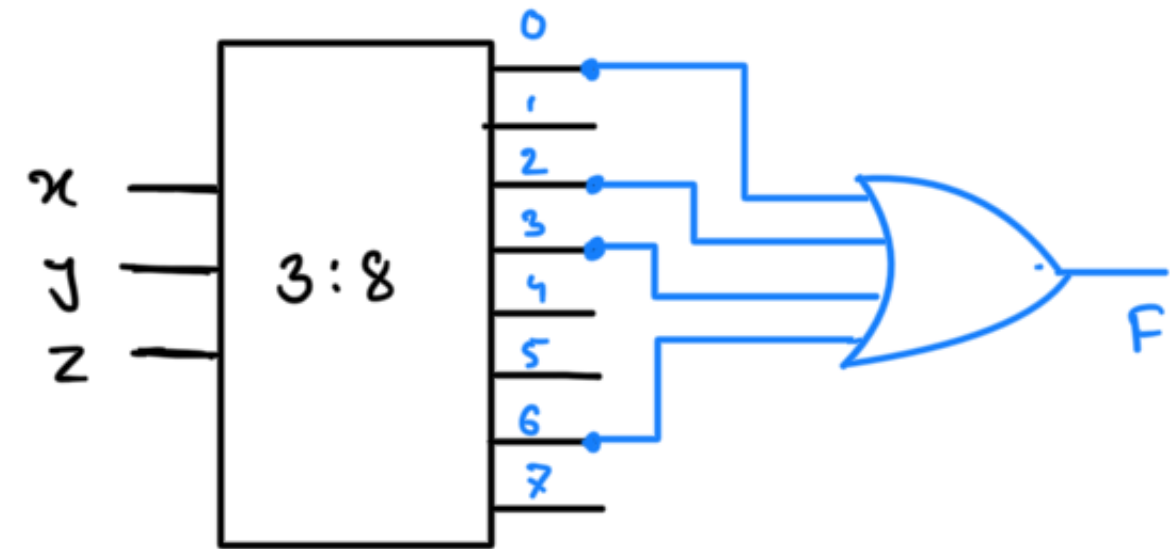
①

$F = \Sigma(0, 2, 3, 6) \rightarrow 8 \text{ outputs, } 3 \text{ inputs}$

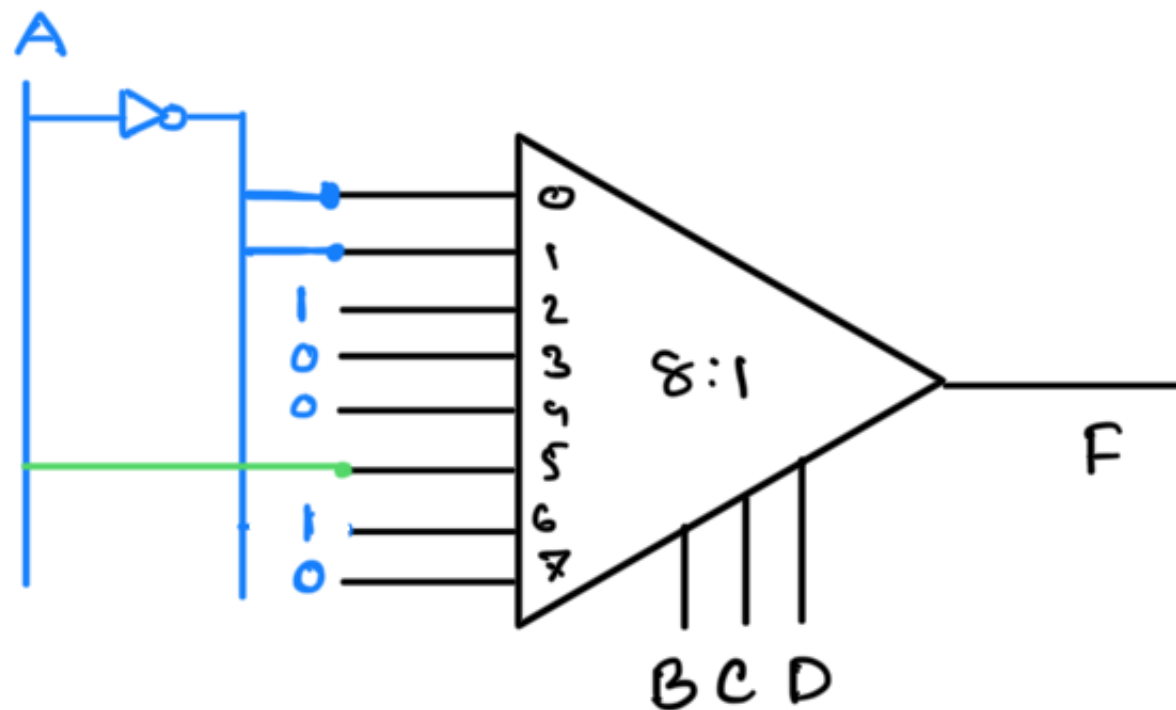
a)



b)

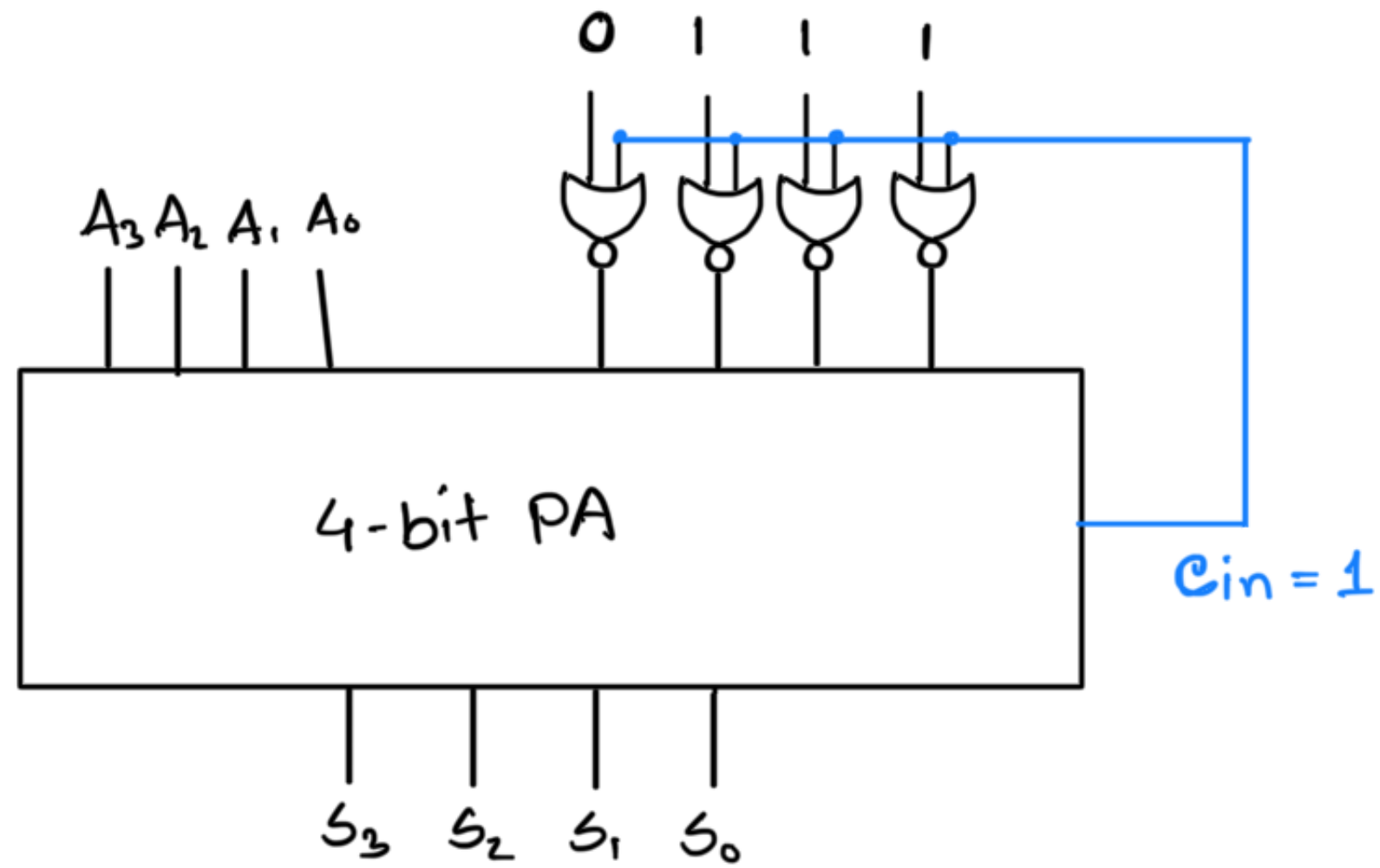


②

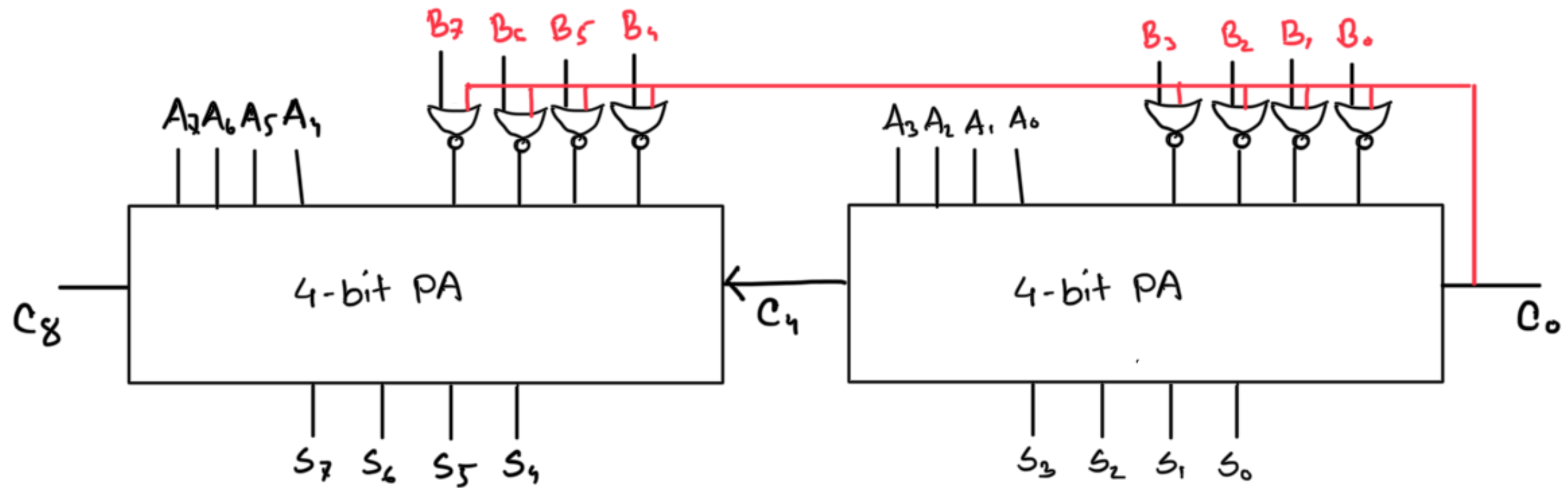


	10	11	12	13	14	15	16	17
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	A'	A'	1	0	0	A	1	0

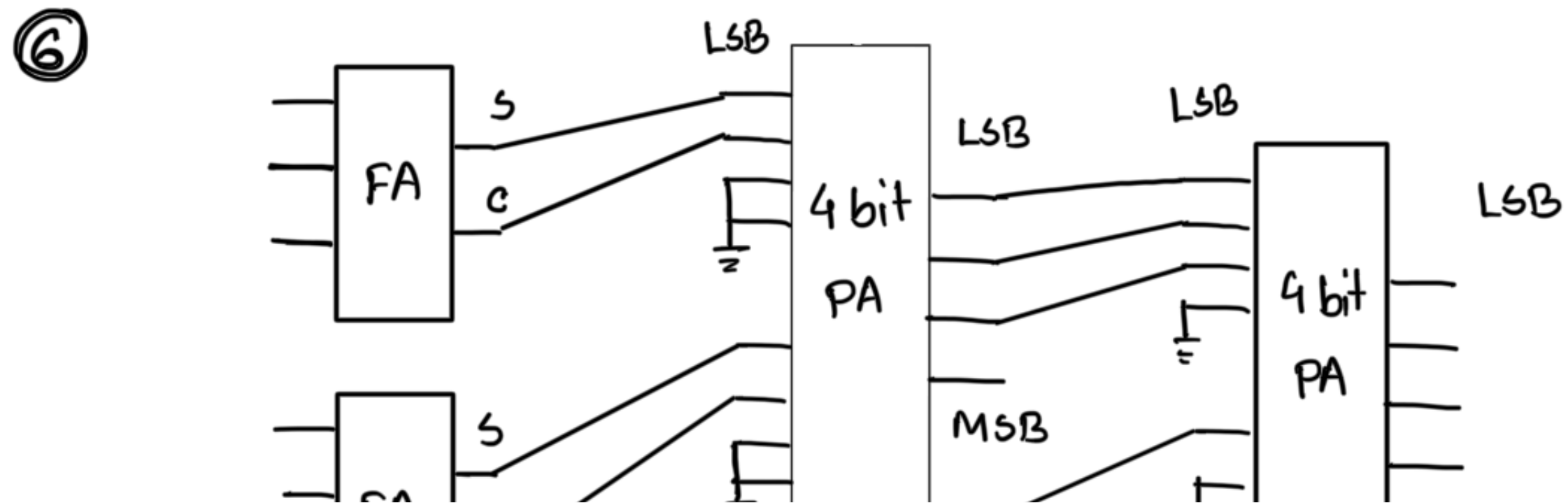
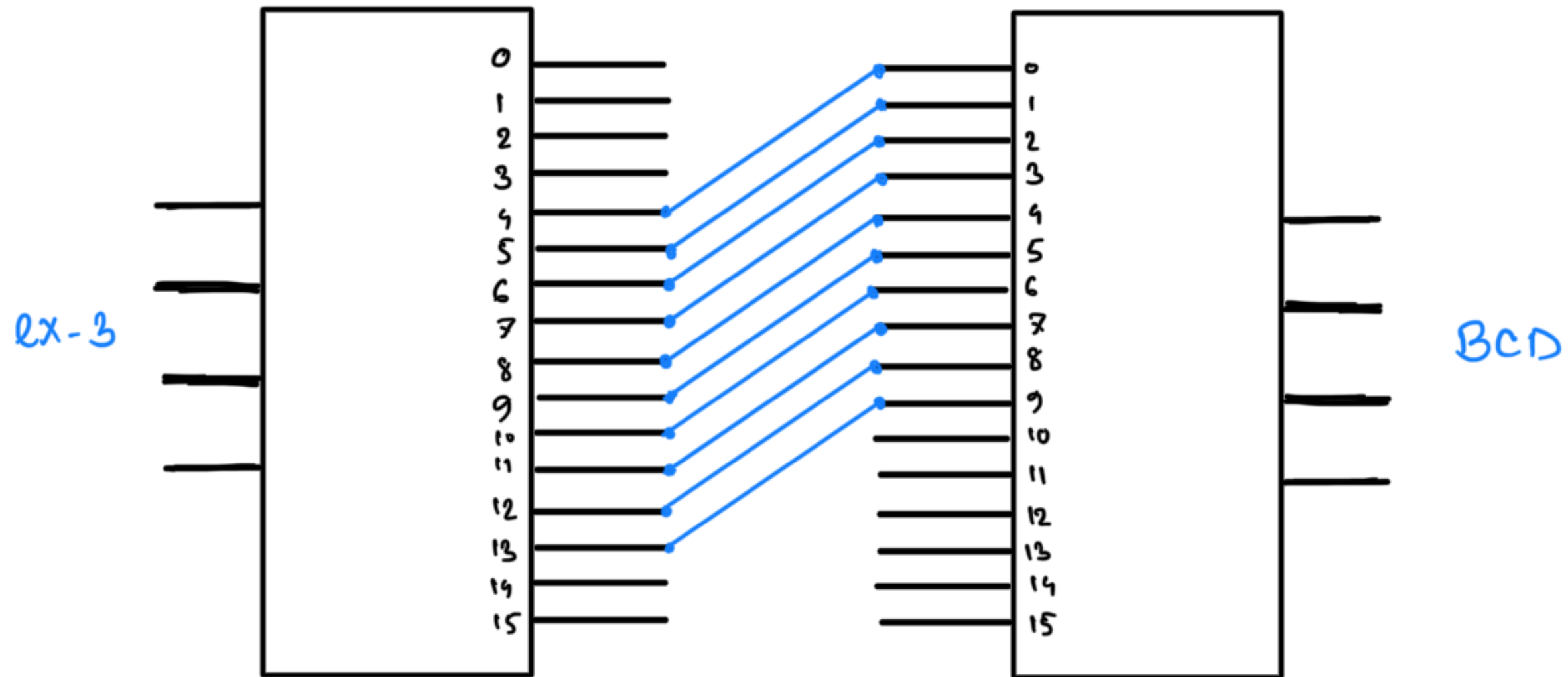
③

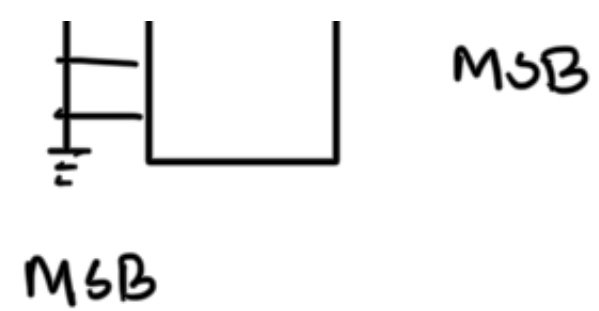
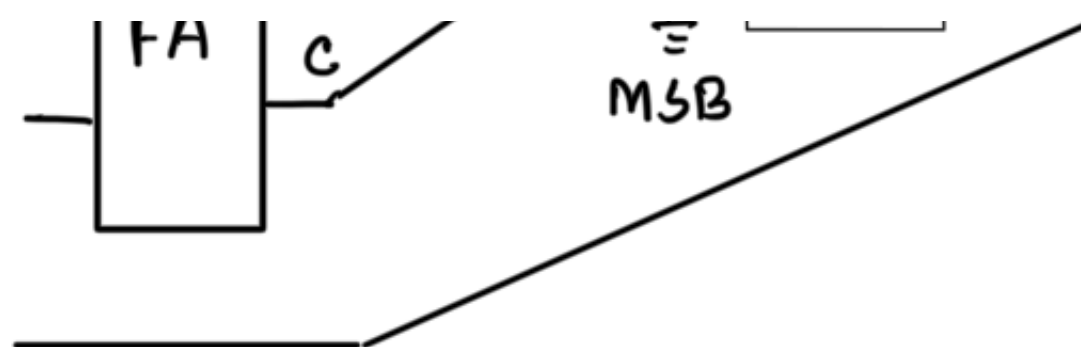


④



⑤ Ex - 4 to BCD converter





⊗

I_3	I_2	I_1	I_0	A_1	A_0
0	0	0	0	1	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	0
0	1	0	0	1	1
0	1	0	1	0	0
0	1	1	0	0	0
0	1	1	1	1	1
1	0	0	0	1	1
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	1	0
1	1	0	0	0	0

A_1

$I_3 I_2$	$I_1 I_0$	00	01	11	10
00		1			1
01		1		1	
11				X	
10		1		1	

$$= I_3' I_2' I_0' + I_3' I_1' I_0' + I_2' I_1' I_0' + I_2 I_1 I_0 + I_3 I_1 I_0$$

A_0

$I_3 I_2$	$I_1 I_0$	00	01	11	10
00		1	1		
01					
11					
10					

$$\begin{array}{cccc|cc}
 1 & 1 & 0 & 1 & 0 & 1 \\
 1 & 1 & 1 & 0 & 0 & 0 \\
 1 & 1 & 1 & 1 & X & X
 \end{array}$$

01	1		1	
11		1	X	
10	1			

$$\begin{aligned}
 = & l_3' l_1' l_0' + l_3' l_2' l_1' + l_2' l_1' l_0' + l_3 l_2 l_0 \\
 & + l_2 l_1 l_0
 \end{aligned}$$